WORKED SOLUTION

STEP III Specimen paper, Question 7

Pure • Reduction of Order

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q7

The key idea

When a second-order linear equation has x^2y'', xy' and y in it, a power of x is the natural thing to try: each term then becomes a multiple of x^α, and the equation turns into a condition on α. Once one solution y_1 is known, writing $y = y_1 v$ is the method of *reduction of order*: the terms in v itself cancel, leaving a first-order equation for v'.

Solution

A power solution

Try $y = x^\alpha$ (for $x > 0$, so that x^α makes sense for any real α). Then $y' = \alpha x^{\alpha-1}$ and $y'' = \alpha(\alpha-1)x^{\alpha-2}$, so

$$xy' - y = (\alpha-1)x^\alpha, \qquad x^2y'' = \alpha(\alpha-1)x^\alpha.$$

Substituting,

$$x^2y'' + (x-2)(xy'-y) = [\alpha(\alpha-1) + (x-2)(\alpha-1)]x^\alpha = (\alpha-1)(\alpha+x-2)\,x^\alpha.$$

For this to vanish for *all* x we need the bracket to be zero identically. The factor $\alpha + x - 2$ depends on x, so it can't be identically zero; hence $\alpha = 1$.

So $y = x$ is a solution, and you can see it directly: $y'' = 0$ and $xy' - y = x - x = 0$. Any multiple $y = Ax$ is then also a solution, which is the sense in which the solution is "proportional to x^α".

The general solution

Following the hint, put $y = xv$, where v is a function of x. Then

$$y' = v + xv', \qquad y'' = 2v' + xv'',$$

and the combination that appears in the equation simplifies nicely:

$$xy' - y = xv + x^2v' - xv = x^2v'.$$

Substituting into the equation,

$$\begin{aligned} 0 &= x^2(2v' + xv'') + (x-2)x^2v' \\ &= x^2(xv'' + 2v' + xv' - 2v') \\ &= x^3\,(v'' + v'). \end{aligned}$$

Ask yourself Why has every term in v itself (with no derivative) disappeared?

Because $y = x$ is a solution: if v were constant, $y = xv$ would be a multiple of x and the left-hand side would be zero. So only v' and v'' can survive. This is what makes reduction of order work, and it is a useful check on the algebra.

On any interval not containing 0 we may divide by x^3, giving

$$v'' + v' = 0.$$

This is first order in $w = v'$: $w' = -w$, so $w = Be^{-x}$ for a constant B. Integrating once more (and absorbing the sign into the constant),

$$v = A + Be^{-x}.$$

Therefore the general solution is

$$\boxed{y = Ax + Bxe^{-x}}$$

with A and B arbitrary constants. It contains two arbitrary constants, as the general solution of a second-order linear equation must, and the two pieces x and xe^{-x} are not multiples of each other.

As a direct check of the new piece, $y = xe^{-x}$ gives $y' = (1-x)e^{-x}$ and $y'' = (x-2)e^{-x}$, so

$$x^2y'' + (x-2)(xy' - y) = (x-2)e^{-x}[x^2 + x(1-x) - x] = 0.$$

A remark on $x = 0$. The coefficient of y'' vanishes at $x = 0$, which is why we divided by x^3 only away from 0. In principle one could take different constants on $x > 0$ and $x < 0$; but every such function has $y(0) = 0$, $y'(0^{\pm}) = A_{\pm} + B_{\pm}$ and $y''(0^{\pm}) = -2B_{\pm}$, so for y to be twice differentiable at 0 we need $B_+ = B_-$ and then $A_+ = A_-$. So $y = Ax + Bxe^{-x}$ is the general solution on the whole real line too.

Checks

- Substitute both $y = x$ and $y = xe^{-x}$ back into the equation, as above: each gives zero.
- Numerically: solving the equation step by step from $x = 1$ with $y(1) = y_0$, $y'(1) = y_1$, the formula predicts $A = y_1$, $B = (y_0 - y_1)e$. With $y_0 = 2$, $y_1 = 1$ this gives $y(4) = 4 + 4e^{-3} \approx 4.199$, and a fourth-order Runge–Kutta integration agrees to better than 10^{-8} for several random starting values.

Common slips

- Concluding $\alpha = 2 - x$ from the factor $(\alpha + x - 2)$: α must be a constant, so this root is not allowed.
- Making an error in y'' for $y = xv$ (it is $2v' + xv''$, not $v' + xv''$). If the terms in v don't cancel, the differentiation has gone wrong.
- Stopping at $v' = Be^{-x}$ and forgetting the second constant when integrating to get v.

Beyond STEP

Reduction of order works for any second-order linear equation $y'' + p(x)y' + q(x)y = 0$: given one solution y_1, the substitution $y = y_1 v$ always gives a first-order linear equation for v', and the second solution can be written as an integral involving y_1 and p. It is closely tied to the Wronskian and Abel's formula, and the trial of x^α at a point like $x = 0$ is the first step of the Frobenius method for series solutions. All of this is covered in Boyce and DiPrima, *Elementary Differential Equations*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.